

LIFESTYLE AND SLEEP QUALITY IN COLLEGE STUDENTS

SAVANNA KERBY
and
ANZLEI LARSEN
and
LAUREN WHITE

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Abstract

This report investigates the primary predictors of sleep quality among undergraduate students ($N = 253$) to determine the relative predictive power of psychological distress versus academic behaviors. Utilizing a dataset containing academic, psychological, and circadian metrics, we employed LASSO regularization and BIC variable selection to identify a parsimonious model that explains approximately 35% of the variance in sleep quality. Our analysis focused on identifying key behavioral and psychological markers that characterize sleep disruption. Notable findings include the significant predictive power of anxiety status and that students with a "normal" anxiety status were forecasted to have sleep disruption scores 1.32 points lower than those with moderate anxiety on a 1 to 18 scale. Additionally, academic engagement, measured by classes missed, emerged as a reliable behavioral marker for sleep health. These results suggest that while sleep is a complex physiological response, it is highly associated with identifiable and modifiable psychological and behavioral patterns, providing a robust framework for anticipating sleep quality trends in the university population.

1 Introduction

Sleep deprivation among college students is widely recognized; however, the underlying causes of poor sleep quality remain an area of ongoing discussion in campus wellness research. It is commonly assumed that sleep quality is driven by academic scheduling, such as early morning classes and demanding workloads. While these factors are relevant, our research suggests that psychological factors and biological predispositions may have a more influence than a student’s schedule alone.

Our analysis is organized around three primary domains:

Academic and Behavioral: Grade Point Average (GPA), number of early classes, and the frequency of missed classes.

Psychological Health: Assessment scores for depression, anxiety, and general stress levels.

Circadian and Sleep Metrics: Self-identified sleep types (Lark, Owl, or Neither), along with average sleep duration on both weekdays and weekends.

Before modeling, we cleaned the data to ensure proper variable types; categorical predictors were treated as factors to allow for the comparison of distinct groups against the “Moderate” anxiety baseline.

1.1 Data Description

This analysis uses the SleepStudy dataset, originally derived from a 2012 study by the Department of Psychology at the University of St. Lawrence (available in the SAGE library under the title, “Learn to Create Bar Charts in RStudio With Data From a Study of Sleep Patterns for College Students (2012)”) on college student sleep patterns and academic performance. The dataset contains 253 undergraduate observations and includes 27 variables capturing a wide range of behavioral, psychological, and academic characteristics.

The variables encompass multiple domains relevant to sleep research, including academic factors, psychological measures, and biological or behavioral indicators. Sleep quality is measured directly through a standardized indicator, allowing for comparisons across individuals. This structure makes the dataset particularly well-suited for examining how environmental, psychological, and biological factors jointly influence sleep health among college students.

Table 1 lists the variables that were the focus of our main analysis efforts, along with their scales.

Variable	Measurement Description	Scale/Range
Poor Sleep Quality	Response: Degree of disruption	[1, 18] (18 = worst)
Anxiety Status	1=Moderate, 2=Normal, 3=Severe	{1, 2, 3}
Depression Score	Degree of depression	[0, 35]
Happiness Score	Degree of happiness	[0, 35]
Classes Missed	Semester absences	[0, 20]
Weekday Sleep	Average hours of nightly sleep	[0, 12+]
Weekday Bed	Bedtime (24.0 = Midnight)	[0, 24]
Lark Owl	1=Lark, 2=Neither, 3=Owl	{1, 2, 3}
Cognition Z-score	Standardized cognitive test score	[-1.96, 1.96]

Table 1: Summary of variables included in the regression model, including units and measurement scales.

2 Model Selection and Validation

When we first began exploring this dataset, we were interested in identifying how various behavioral and psychological variables were associated with both sleep quality and cognitive performance. We initially fit multiple linear regression models using *CognitionZScore* as the response variable, testing predictors that were either correlated with the score or theoretically associated with cognitive performance. However, these models exhibited very low explanatory power, with an R^2 of approximately 0.06.

In contrast, when modeling *PoorSleepQuality* as the response variable, we observed substantially higher explanatory power, with R^2 values ranging from 0.20 to 0.30. Given that our primary interest is student well-being, we refined our research question to focus specifically on the factors that characterize sleep quality. In this report, we seek to identify the primary predictors of sleep disruption and determine which behavioral markers are most strongly associated with sleep outcomes.

To identify relevant predictors, we applied LASSO regression, which performs variable selection by shrinking less important coefficients toward zero. This allowed us to reduce the number of candidate predictors while retaining those with the strongest predictive relationships to *PoorSleepQuality*. The variables selected through LASSO included: *LarkOwl*, *NumEarlyClass*, *ClassesMissed*, *DepressionScore*, *AnxietyScore*, *AnxietyStatus*, *Happiness*, *WeekdayBed*, and *WeekdaySleep*.

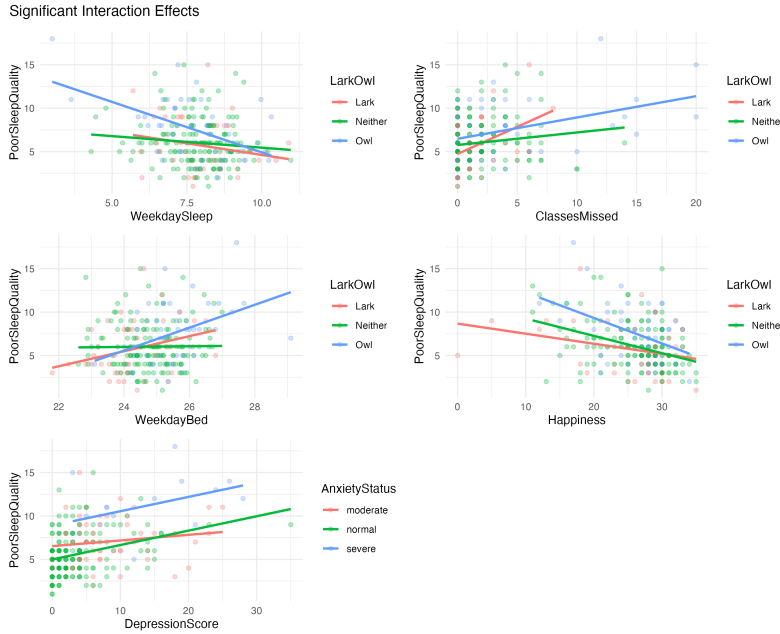


Figure 1: Plotting the significant interactions between the variables.

Using this reduced set of predictors, along with several interaction terms motivated by our exploratory data analysis (see Figure 1), we performed best subset selection. We evaluated candidate models using three criteria: maximized adjusted R^2 , minimized Mallows' C_p , and minimized Bayesian Information Criterion (BIC). These metrics allowed us to compare model fit while penalizing for excessive model complexity.

To assess out-of-sample performance, we conducted ten-fold cross-validation. For each candidate model, we computed the root mean squared error (RMSE) to measure predictive

accuracy on unseen data. The models that maximized adjusted R^2 and minimized C_p coincided, achieving the lowest RMSE of 2.3915 and an R^2 of 0.3339.

We selected model (1) as our final predictive model because it balances interpretability, predictive performance, and consistency with the variables identified through LASSO. The final model includes *LarkOwl*, *ClassesMissed*, *AnxietyStatus*, *Happiness*, *WeekdaySleep*, *WeekdayBed*, *DepressionScore*, and several interaction terms. This selection provides a comprehensive view of how behavioral, psychological, and circadian factors jointly influence poor sleep quality.

$$\begin{aligned}
\text{PoorSleepQuality}_i = & \beta_0 + \beta_1 \times \text{I}(\text{LarkOwl}_i = \text{Neither}) \\
& + \beta_2 \times \text{I}(\text{LarkOwl}_i = \text{Owl}) \\
& + \beta_3 \times \text{ClassesMissed}_i \\
& + \beta_4 \times \text{I}(\text{AnxietyStatus}_i = \text{normal}) \\
& + \beta_5 \times \text{I}(\text{AnxietyStatus}_i = \text{severe}) \\
& + \beta_6 \times \text{DepressionScore}_i \\
& + \beta_7 \times \text{WeekdaySleep}_i \\
& + \beta_8 \times \text{WeekdayBed}_i \\
& + \beta_9 \times \text{Happiness}_i \\
& + \beta_{10} \times (\text{DepressionScore}_i \times \text{I}(\text{AnxietyStatus}_i = \text{normal})) \\
& + \beta_{11} \times (\text{DepressionScore}_i \times \text{I}(\text{AnxietyStatus}_i = \text{severe})) \\
& + \beta_{12} \times (\text{I}(\text{LarkOwl}_i = \text{Neither}) \times \text{WeekdayBed}_i) \\
& + \beta_{13} \times (\text{I}(\text{LarkOwl}_i = \text{Owl}) \times \text{WeekdayBed}_i) \\
& + \beta_{14} \times (\text{I}(\text{LarkOwl}_i = \text{Neither}) \times \text{Happiness}_i) \\
& + \beta_{15} \times (\text{I}(\text{LarkOwl}_i = \text{Owl}) \times \text{Happiness}_i) + \epsilon_i,
\end{aligned} \tag{1}$$

where

$$\epsilon_i \sim N(0, \sigma^2).$$

After selecting the final model, we evaluated the standard assumptions of multiple linear regression to determine whether any transformations or adjustments were necessary. From the partial regression plots (see Figures 5 & 6), we observed no clear violations of linearity, as the relationships between the predictors and the response appeared approximately linear. To assess homoscedasticity, we examined the residuals versus fitted values plot (see Figure 9) and found no evident patterns or funneling, suggesting constant variance of the errors.

The normality of the residuals was evaluated using a Q-Q plot (see Figure 7) and a histogram of residuals (see Figure 8); both indicated that the residuals were approximately normally distributed. Independence of errors was assumed based on the study design and the lack of temporal or clustered structure in the data. Additionally, we assessed the presence of influential observations using Cook's distance (see Figure 10) and leverage diagnostics (see Figure 11) and found no observations exceeding conventional thresholds for undue influence on the fitted model.

Finally, multicollinearity was assessed using Variance Inflation Factors (VIF). VIF values from a main-effects-only model were all close to 1, indicating no meaningful multicollinearity among the predictors at baseline. When interaction terms were included, VIF values increased for some predictors—particularly those involving *LarkOwl*—but this inflation is expected due to the structural dependence introduced by interaction terms rather

than indicating problematic redundancy. Importantly, none of the values reached levels typically associated with severe multicollinearity. Overall, these diagnostics suggest that the assumptions of multiple linear regression were reasonably satisfied, and no additional transformations were required.

2.1 Note on Substance Use and Predictive Utility

Initial exploratory analysis and variable screening also accounted for Alcohol Use within the student population. Despite the common assumption that alcohol consumption significantly alters sleep architecture, it did not emerge as a significant predictor of poor sleep quality within this specific dataset.

3 Analyses, Results, and Interpretation

We fit Model 1 using ordinary least squares. The baseline categories for the categorical variables in this model were $LarkOwl = Lark$ (the morning lark chronotype) and $AnxietyStatus = moderate$ (moderate anxiety). The regression summary output from R can be found in Table 2.

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-1.9963	9.8304	-0.20	0.8393	
I(LarkOwl _i = Neither)	14.1312	10.2640	1.38	0.1699	
I(LarkOwl _i = Owl)	-7.5826	12.0508	-0.63	0.5298	
Classes Missed	0.1008	0.0502	2.01	0.0456	*
I(Anxiety Status _i = normal)	-1.3240	0.5897	-2.25	0.0257	*
I(Anxiety Status _i = severe)	0.9617	1.2430	0.77	0.4399	
Depression Score	0.0290	0.0549	0.53	0.5977	
Weekday Sleep	-0.2161	0.1592	-1.36	0.1759	
Weekday Bed	0.4288	0.3714	1.15	0.2494	
Happiness	-0.0251	0.0529	-0.47	0.6354	
Depression Score × I(Anxiety Status _i = normal)	0.0791	0.0668	1.18	0.2378	
Depression Score × I(Anxiety Status _i = severe)	0.1369	0.0895	1.53	0.1272	
I(LarkOwl _i = Neither) × WeekdayBed _i	-0.5176	0.4072	-1.27	0.2049	
I(LarkOwl _i = Owl) × WeekdayBed _i	0.5049	0.4772	1.06	0.2912	
I(LarkOwl _i = Neither) × Happiness _i	-0.0487	0.0629	-0.77	0.4394	
I(LarkOwl _i = Owl) × Happiness _i	-0.1733	0.0885	-1.96	0.0515	.

Table 2: Regression summary output for the model shown in Eq. (1).

The fitted model is thus:

$$\begin{aligned}
\widehat{\text{PoorSleepQuality}}_i = & -2.00 + 14.13 \times \text{I}(\text{LarkOwl}_i = \text{Neither}) \\
& - 7.58 \times \text{I}(\text{LarkOwl}_i = \text{Owl}) \\
& + 0.10 \times \text{ClassesMissed}_i \\
& - 1.32 \times \text{I}(\text{AnxietyStatus}_i = \text{normal}) \\
& + 0.96 \times \text{I}(\text{AnxietyStatus}_i = \text{severe}) \\
& + 0.03 \times \text{DepressionScore}_i \\
& - 0.22 \times \text{WeekdaySleep}_i \\
& + 0.43 \times \text{WeekdayBed}_i \\
& - 0.03 \times \text{Happiness}_i \\
& + 0.08 \times (\text{DepressionScore}_i \times \text{I}(\text{AnxietyStatus}_i = \text{normal})) \\
& + 0.14 \times (\text{DepressionScore}_i \times \text{I}(\text{AnxietyStatus}_i = \text{severe})) \\
& - 0.52 \times (\text{I}(\text{LarkOwl}_i = \text{Neither}) \times \text{WeekdayBed}_i) \\
& + 0.50 \times (\text{I}(\text{LarkOwl}_i = \text{Owl}) \times \text{WeekdayBed}_i) \\
& - 0.05 \times (\text{I}(\text{LarkOwl}_i = \text{Neither}) \times \text{Happiness}_i) \\
& - 0.17 \times (\text{I}(\text{LarkOwl}_i = \text{Owl}) \times \text{Happiness}_i)
\end{aligned} \tag{2}$$

We see that at the $\alpha = 0.05$ significance level, the only explanatory variables with significant effects are *ClassesMissed* and the indicator $\text{I}(\text{AnxietyStatus}_i = \text{normal})$. This was surprising, as none of the interaction effects were significant at the 0.05 level, even though cross-validation had selected models including these interactions as important for prediction.

The coefficient for *ClassesMissed* indicates that for each additional class missed (a one-unit increase in *ClassesMissed*), the expected value for *PoorSleepQuality* increases by 0.1008 on average, holding other variables constant.

The coefficient for $\text{I}(\text{AnxietyStatus}_i = \text{normal})$ is -1.3240 , indicating that when *DepressionScore* = 0, students with normal anxiety have, on average, 1.3240 lower *PoorSleepQuality* scores than the reference group (*AnxietyStatus* = *moderate*), holding other variables constant. This coefficient is substantial relative to the 1–18 scale, especially when compared to the smaller effect size of *ClassesMissed*.

The coefficient for $\text{I}(\text{LarkOwl}_i = \text{Owl}) \times \text{Happiness}_i$ suggests a potentially interesting association ($p = 0.0515$), though it is not significant at the 0.05 level. The estimated coefficient indicates that the slope of *Happiness* for night owls is 0.1733 lower than for the reference group (morning larks), holding other variables constant. The estimated relationship between *Happiness* and *PoorSleepQuality* is slightly negative for morning larks and students with neither chronotype, and more negative for night owls. A one-unit increase in *Happiness* for morning larks corresponds with an estimated decrease of 0.0251 in *PoorSleepQuality* on average. For night owls, the corresponding estimated decrease is 0.1984. However, neither the main effect of *Happiness* nor the interaction terms are statistically significant, so the evidence for this pattern is weak.

We noticed that, compared to models we had fit earlier on in the process, this model had very few statistically significant coefficients. This is likely due to the inclusion of multiple interaction terms, which substantially increased model complexity. Partitioning the data across many subgroups effectively reduced the amount of information available for estimating each parameter. As a result, the variance of the coefficient estimates increased.

This inflated the standard errors and widened the confidence intervals. These confidence intervals can be seen in Figure 2 in the Appendix. Note the especially wide intervals for the *LarkOwl* variable, which dwarf the other intervals.

We wanted to make sure that the interaction terms were important. So, we ran a partial F-test comparing the full model, Model (1), to a model with the same variables but without the interaction terms. This resulted in a p -value of 0.06955, indicating that the interaction terms do not significantly improve model fit when focusing on our ability to conduct inference.

However, because our primary goal is prediction, we ran another 10-fold cross-validation with the two models, with and without interactions. The model including interactions resulted in a slightly lower RMSE (2.388) than the model without interactions (2.391). This suggests a marginal predictive advantage to the model with interactions. Hence, we concluded that these interactions were still important for predictive value.

4 Conclusions

Our analysis demonstrates that poor sleep quality among undergraduate students can be effectively modeled through a combination of psychological indicators and behavioral markers. By employing LASSO regularization and BIC selection, we identified a parsimonious model that accounts for approximately 35% of the variance in sleep disruption scores. This suggests that while sleep is a complex physiological response, a significant portion of its variability is predictable using a targeted set of lifestyle and mental health metrics.

The most influential predictors in our model were Anxiety Status and Classes Missed. Students with a “Normal” anxiety status were predicted to have significantly lower disruption scores (an improvement of 1.32 points) compared to those with moderate anxiety. Additionally, the positive coefficient for missed classes indicates that academic disengagement is a reliable behavioral marker for deteriorating sleep health. While factors like *DepressionScore* and *WeekdaySleep* duration did not reach the threshold for individual statistical significance in this specific model, their inclusion was necessary for optimal predictive performance, likely due to their shared variance with other psychological predictors.

Ultimately, these findings suggest that poor sleep quality is not an inevitable byproduct of university life, but rather a multifactorial response associated with identifiable habits and psychological states. From a predictive standpoint, students who maintain consistent class attendance and report lower anxiety levels are forecasted to experience significantly higher sleep quality. While this study identifies associations rather than direct causation, it provides a robust framework for anticipating which students may be at the highest risk for severe sleep disruption.

APPENDIX

5 Documentation of Revision Process

This report was revised with the assistance of Generative AI (Google Gemini) to ensure technical accuracy and professional clarity.

5.1 Revision Goals and Prompt

We provided the AI with our full draft and a specific prompt to:

1. Improve overall grammar and professional flow.
2. Shift the tone from “causal” (e.g., “how to improve sleep”) to “predictive” (e.g., “identifying markers of sleep quality”).
3. Ensure the specific 1–18 scale of the response variable was clearly explained to the reader.

5.2 Primary Recommendations Adopted

The following improvements were made based on the feedback provided:

- **Verbiage Alignment:** Replaced causal terms like “influence” and “effect” with predictive terms such as “associated with,” “forecasted,” and “predictive markers.”
- **Statistical Clarity:** Explicitly defined the *PoorSleepQuality* scale early in the report to clarify that higher scores indicate worse sleep.
- **Structural Documentation:** Created a summary table of variable scales and ranges to provide the reader with a quick reference for the non-standardized units.
- **Baseline Correction:** Explicitly stated the reference groups (Moderate Anxiety and Morning Larks) to ensure the regression coefficients were interpretable.

Most of the suggestions were for wording, and we implemented the fixes where indicated.

6 Figures

Below are assorted figures illustrating the exploratory data analysis and linear regression assumption checking we performed.

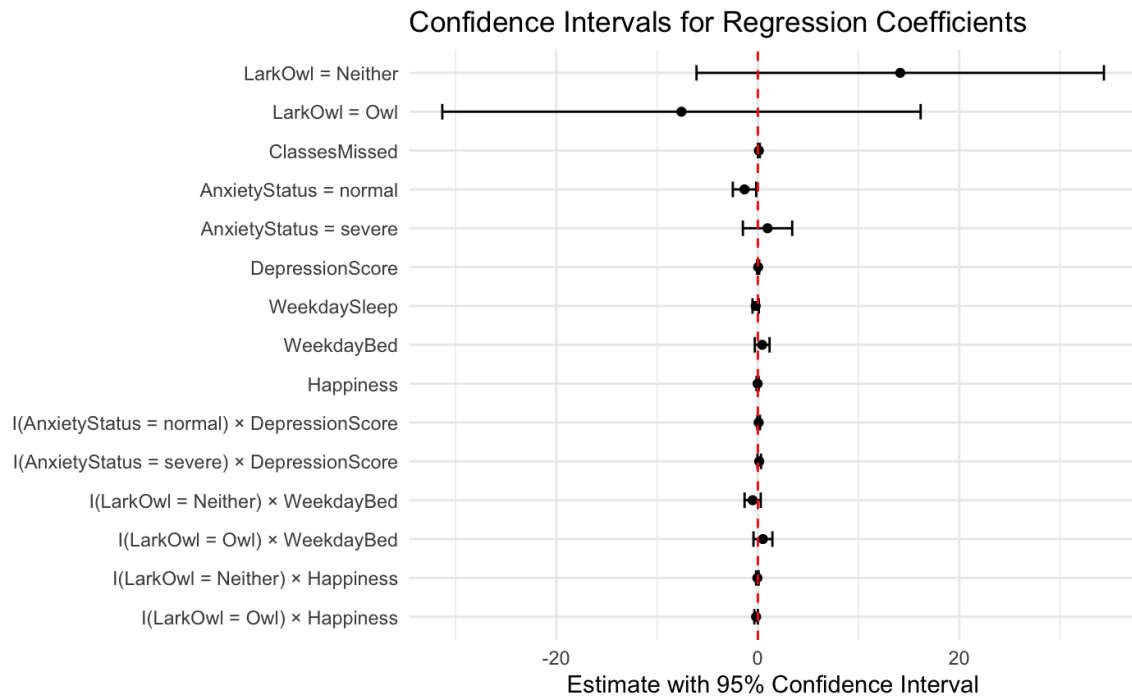


Figure 2: 95% confidence intervals for our fitted model.

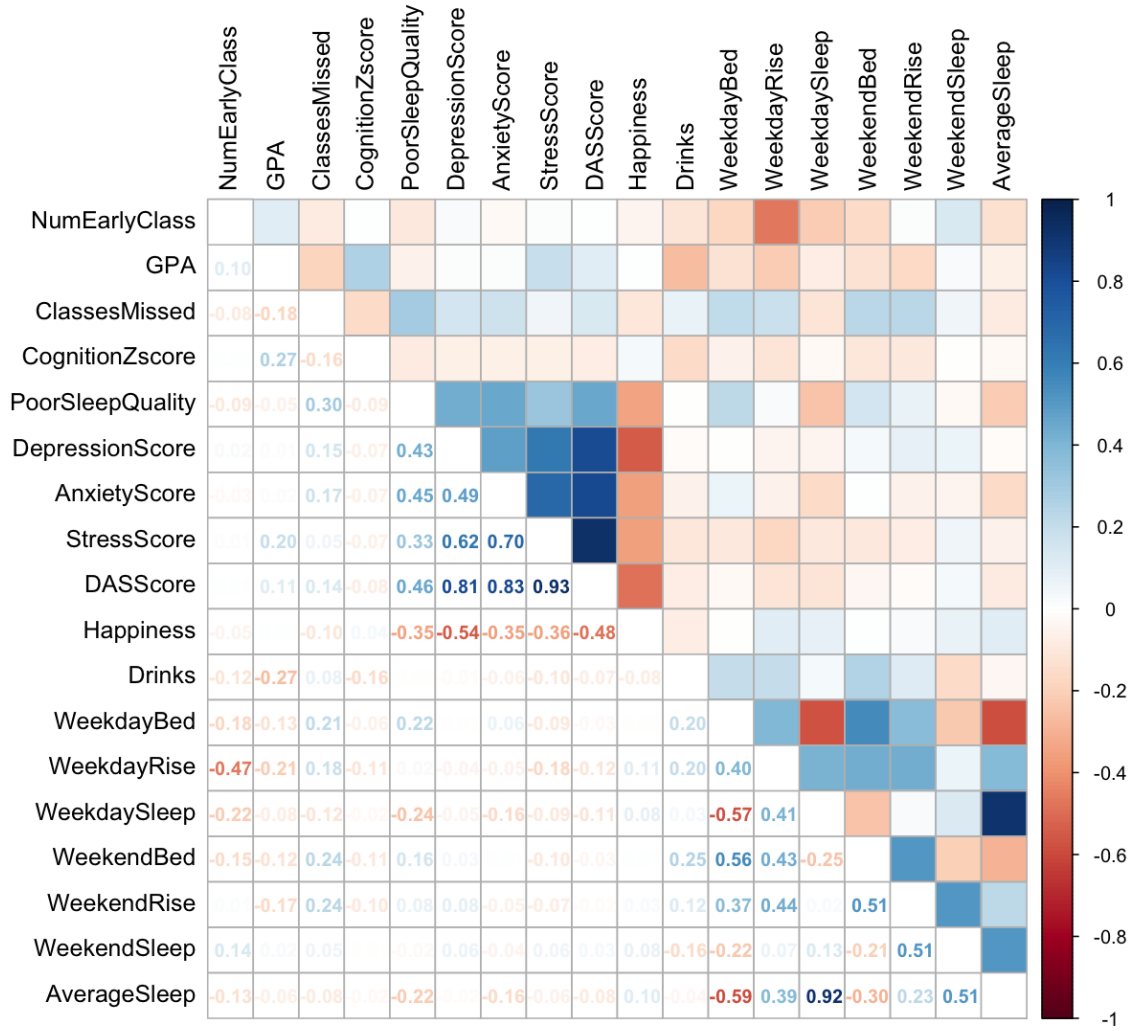


Figure 3: Correlation Matrix of every numeric variable in the dataset.

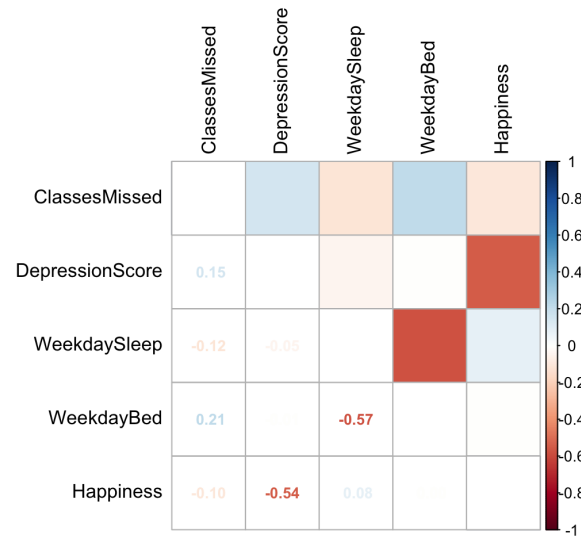


Figure 4: Correlation Matrix of the numeric variables in our model.

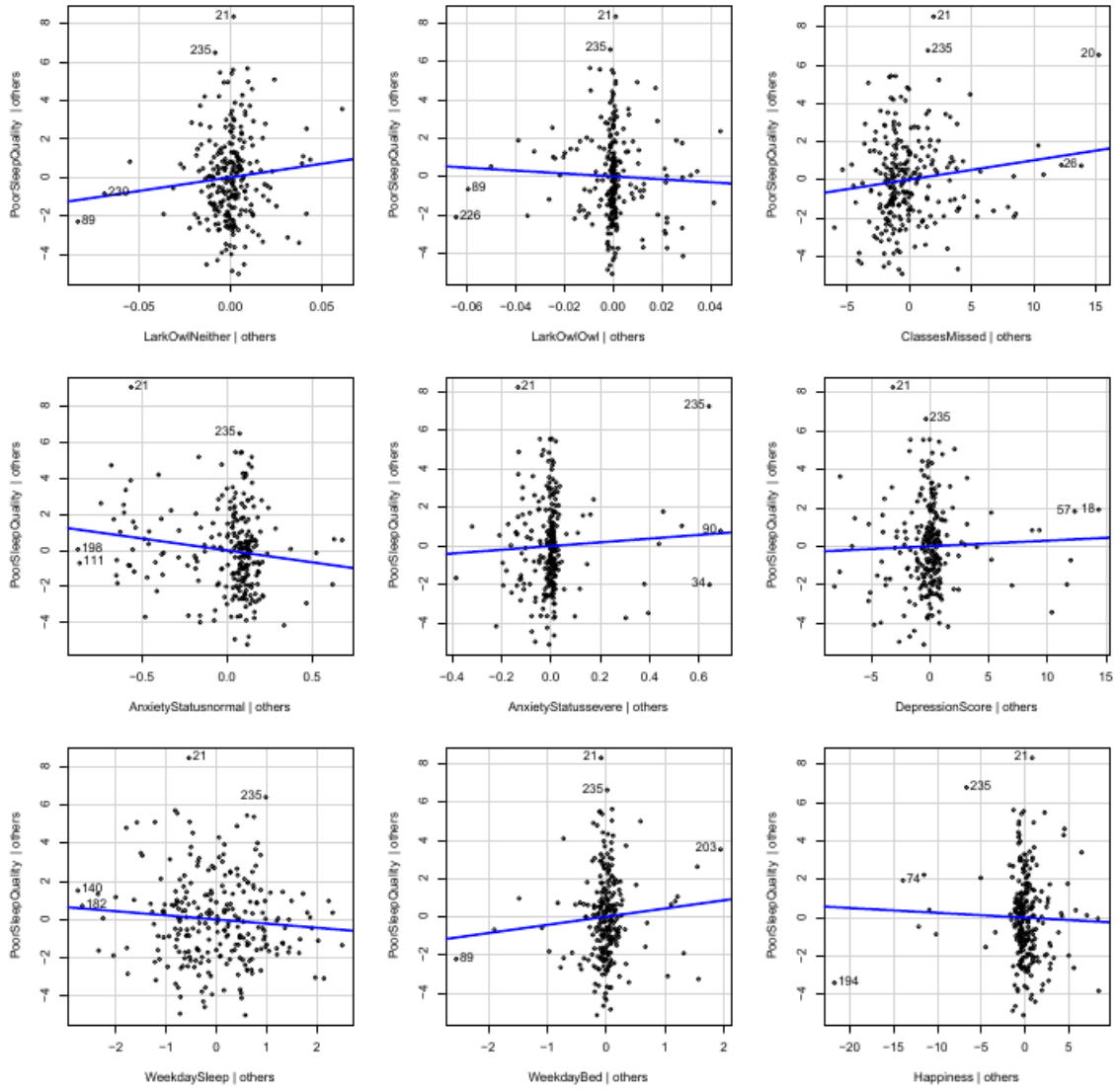


Figure 5: Partial Regression Plots 1

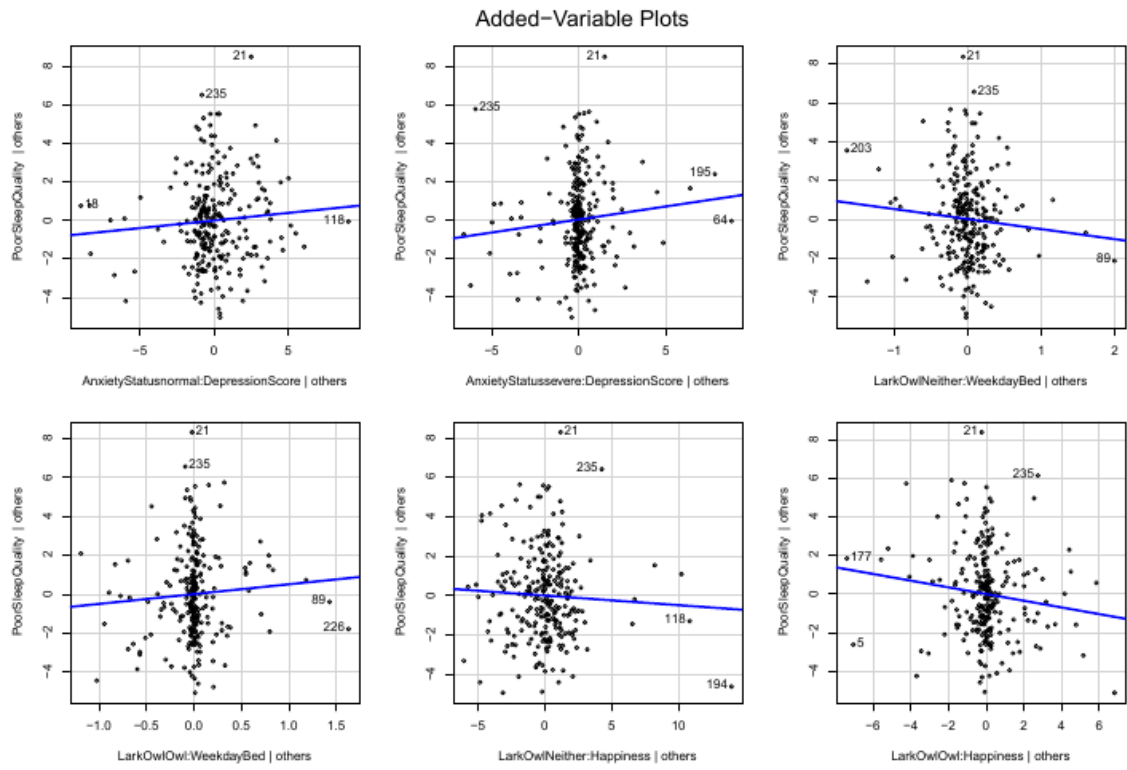


Figure 6: Partial Regression Plots 2

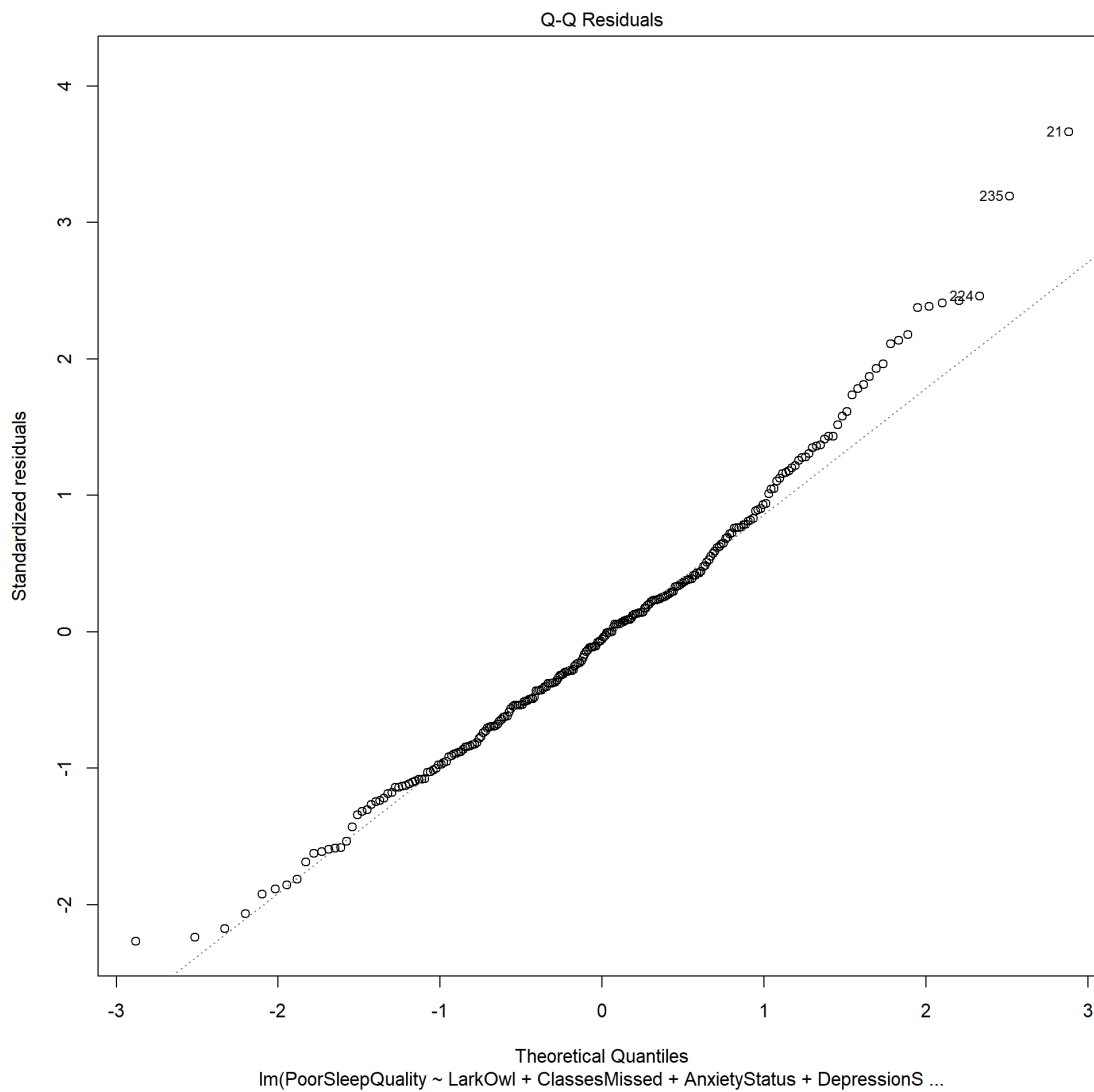


Figure 7: Q-Q Plot

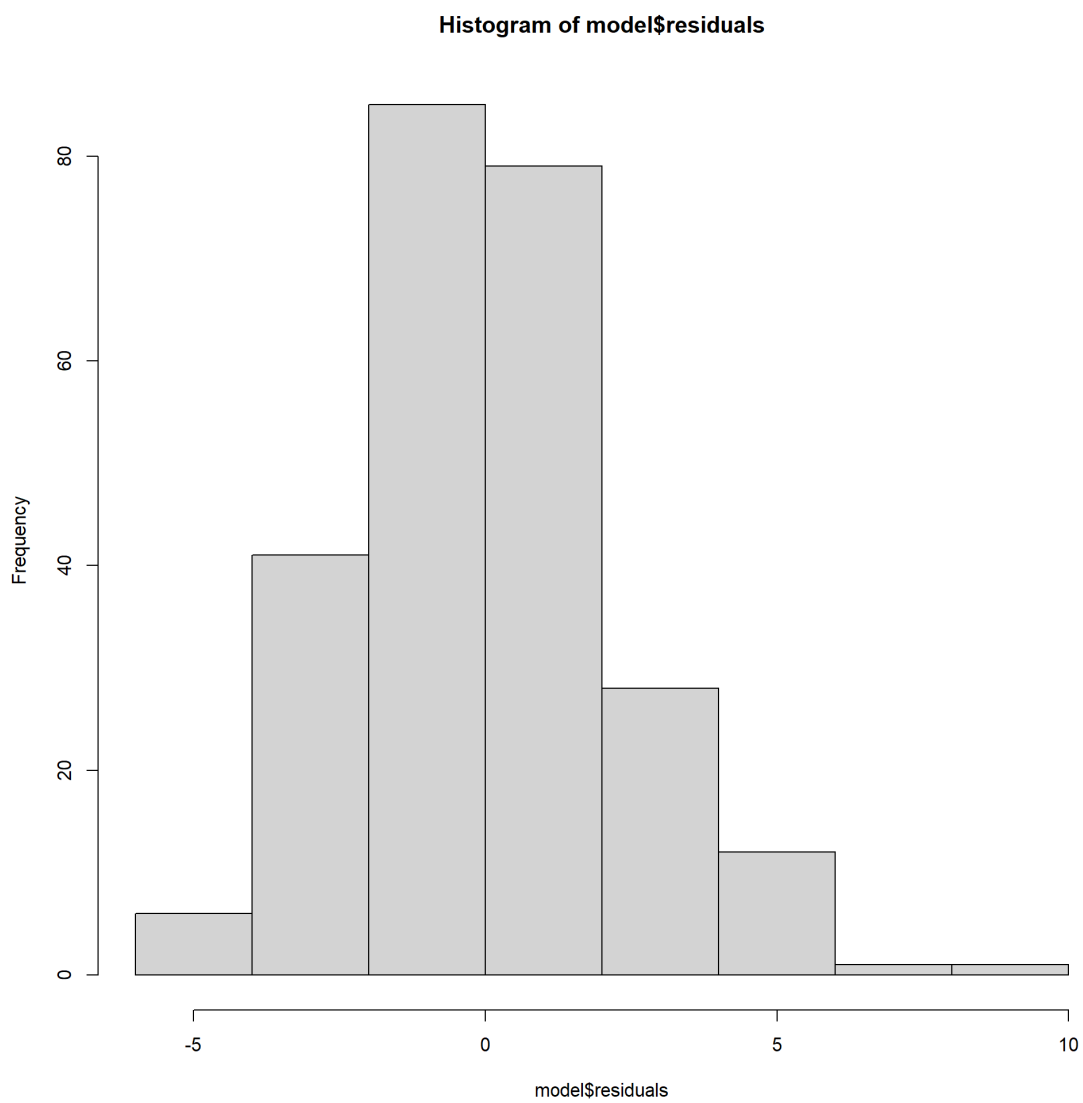


Figure 8: Histogram of Residuals

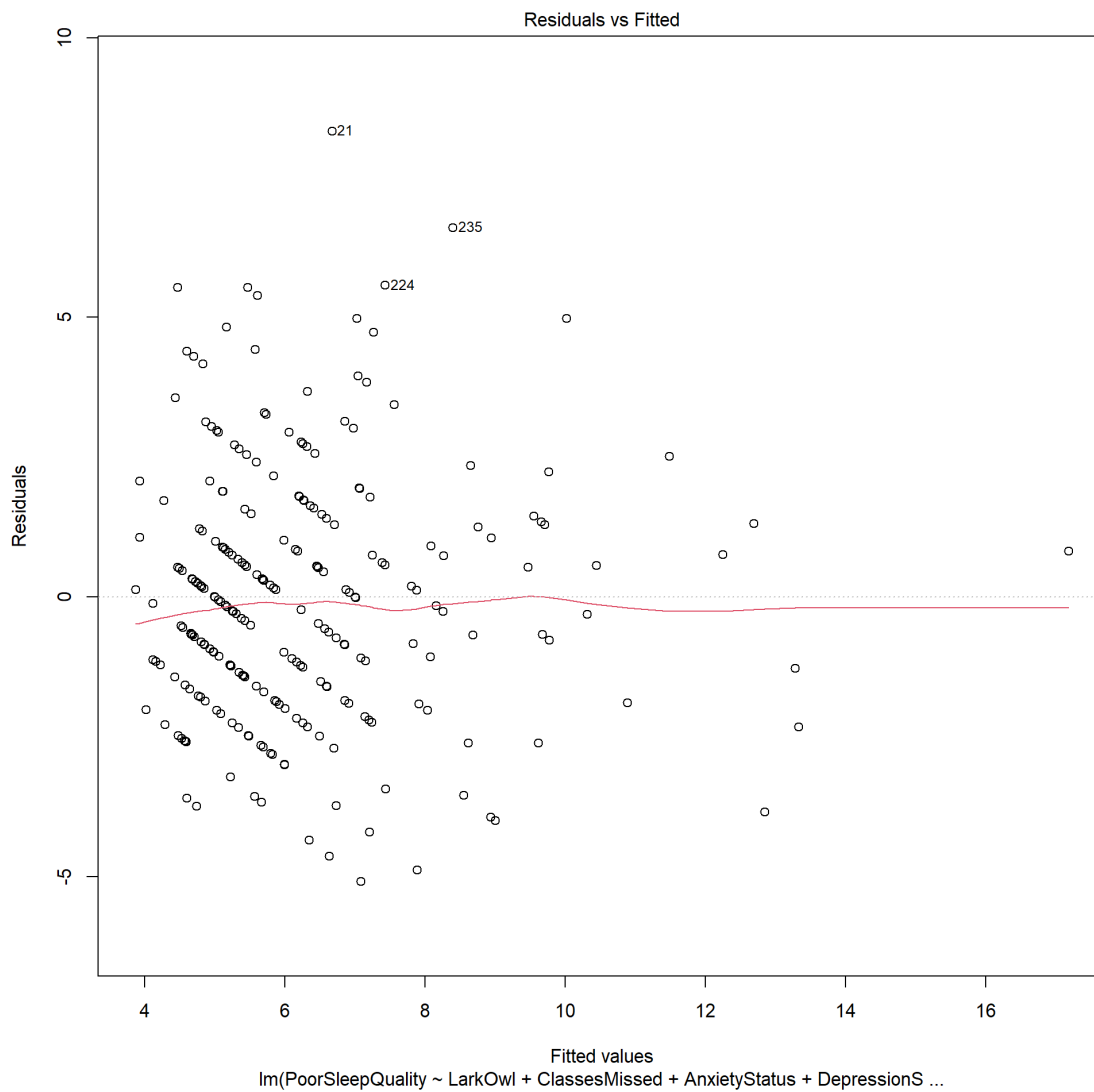


Figure 9: Fitted Values Versus Residuals Plot

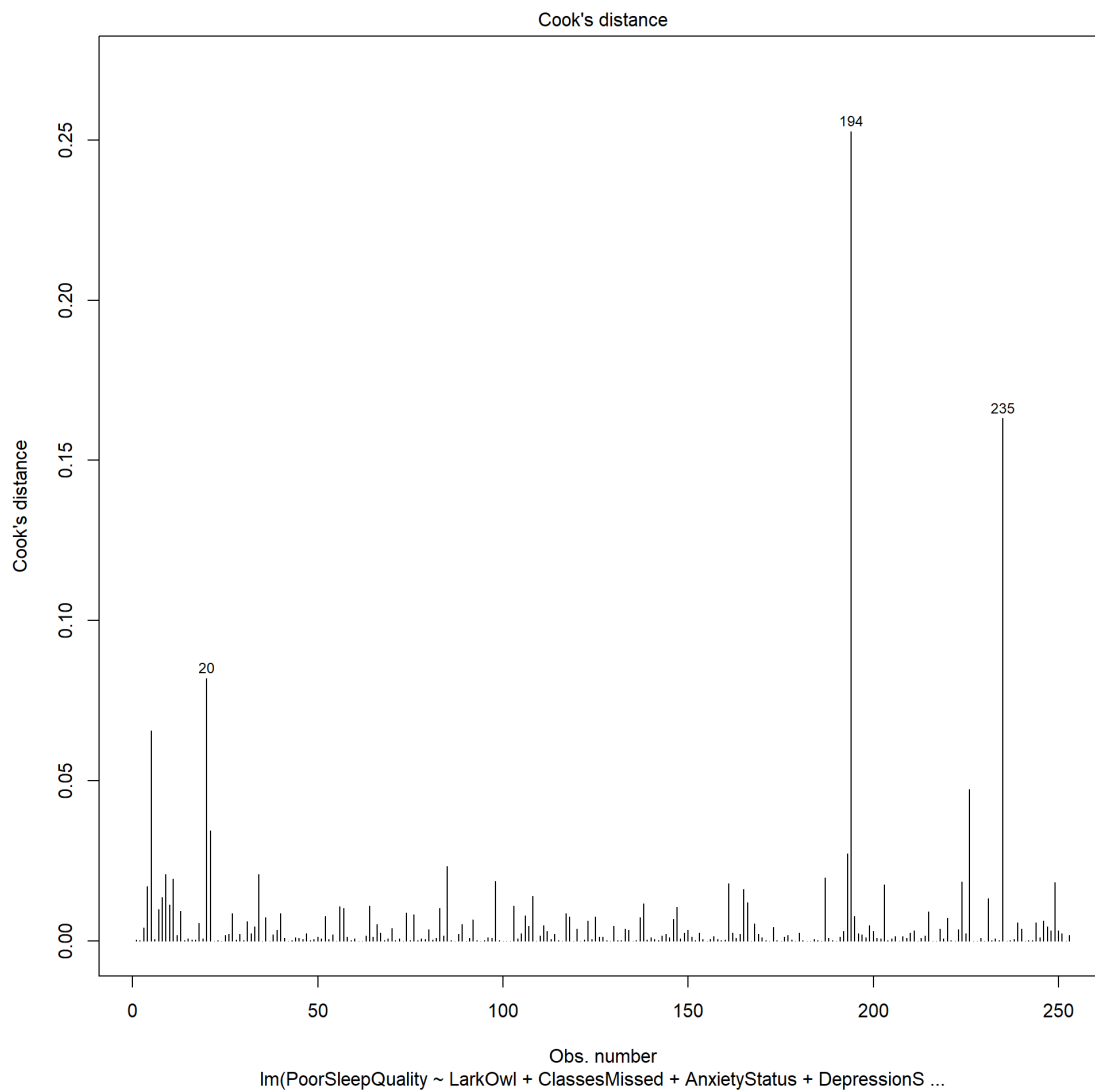


Figure 10: Cook's Distance Plot

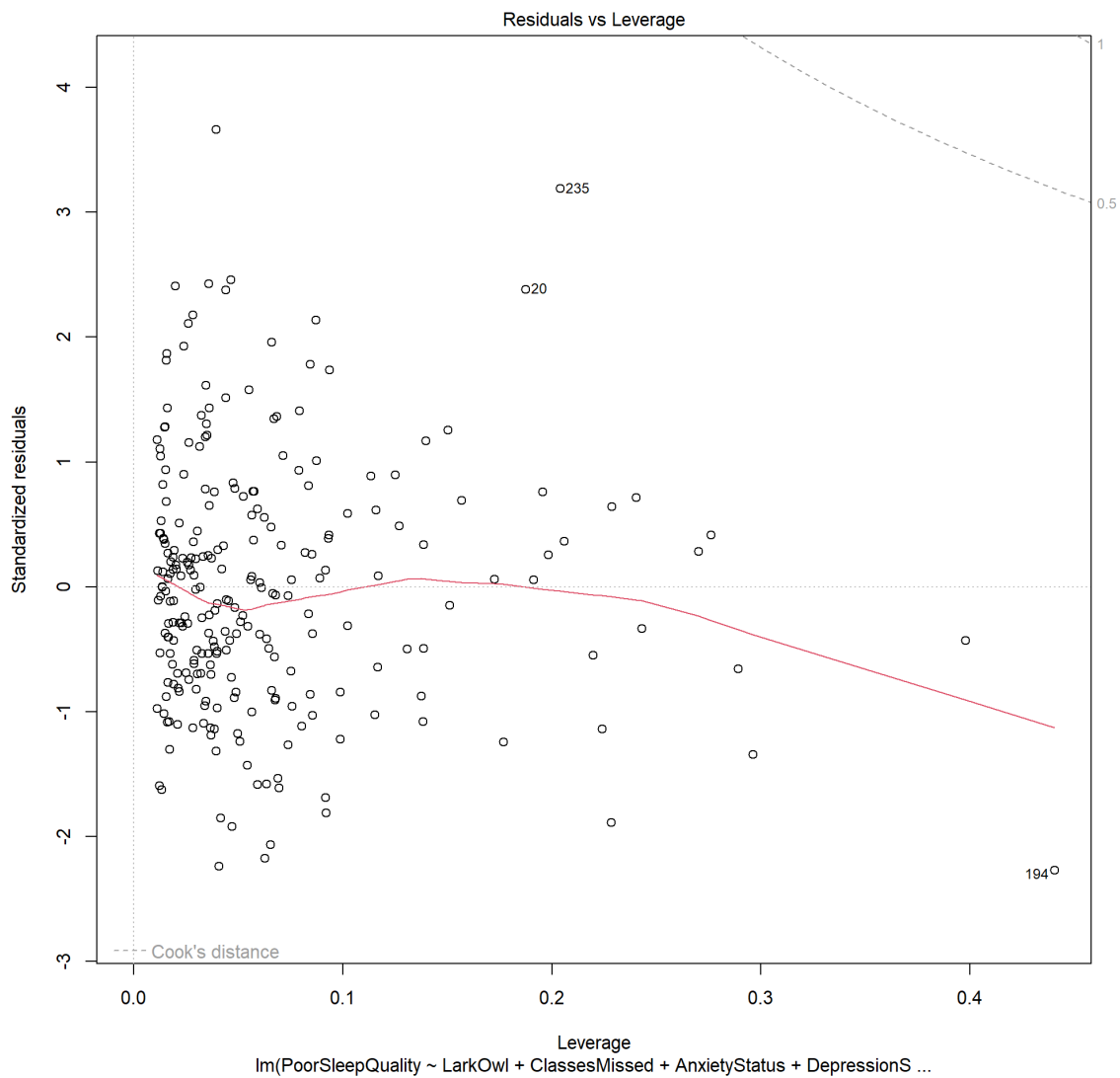


Figure 11: Residuals Versus Leverage Plot